

$M[2500:2503] \leftarrow 1000$

$M[1000:1003] = \text{Integer Variable.}$

$M[2500:2503] = \text{Pointer variable}$

$2500 \rightarrow \text{Base Address of pointer itself.}$

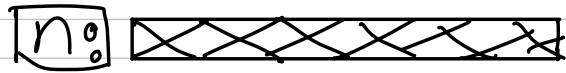
"Every Address in Application is a positive integer"

Though it is possible to store address in an unsigned integer, we do not it.

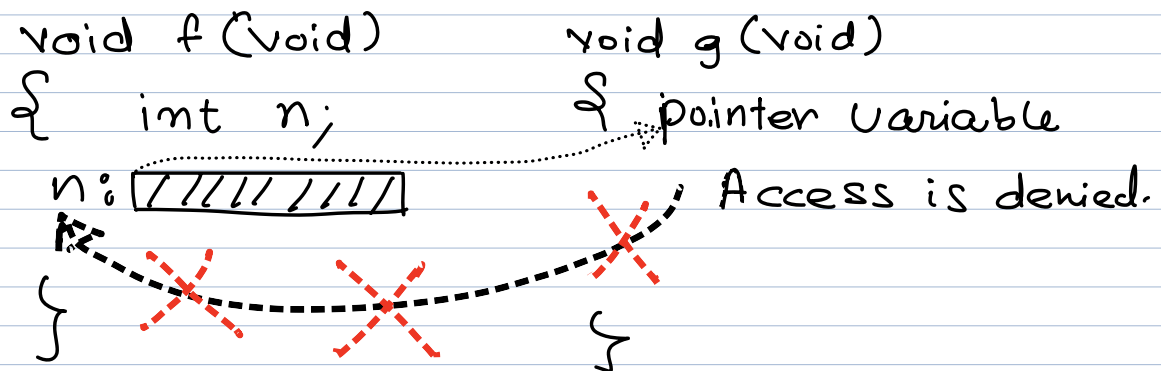
Why? Because we would not be able to tell when an unsigned integer is storing a data value & when it is storing an address.

Therefore a special type of variable is needed to store addresses. And this requirement is fulfilled by pointer variable.

- 1) We are interested in accessing allocated memory locations. Using a variable name is one of the ways to do it.



- 2) Not all memory names are accessible from everywhere. e.g. a variable defined 'INSIDE' one function is not accessible in any other function.



- 3) Not all allocated memory blocks have names associated with them.
∴ pointers are only way to access such memory blocks.

```
int n;  
struct debit_card my_db_card;
```

int n; n type int hai.

If you want to make a particular variable array type.

int n[1];

int n[5]

int [5]
int n away 5 ko

int n=1000;

int n;

int *p



char *p1;

short *p2;

unsigned int *p3;

float *p4;

double *p5;

struct Date *pDate;

long long int *p6;

int *ptr; ptr hai int type ka memori address
base addr. Store karunga uska dekh
pointer variable hai.

`DataTypeName* p;`

`p` हे `DataTypeName` ह्या टाइप च्या मेमरी
एड्रेसचा वेग `Address` store करण्यासाठी
तयार केलेला `pointer variable` आहे.